

Dynamic Programming

An efficient technique that can be used to solve problems that can be divided into smaller problems.

Matrix Multiplication

Given $M_1, M_2, \dots, M_n$, compute the product $M_1 \cdot M_2 \cdot M_3 \dots M_n$, where M_i has dimensions $d_{i-1} \times d_i$.

Objective: Minimize the total scalar multiplications.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}_{2 \times 2} \begin{pmatrix} e \\ f \end{pmatrix}_{2 \times 1} = \begin{pmatrix} ae + bf \\ ce + df \end{pmatrix}_{2 \times 1}$$

$$2 \times 2 \times 1 = 4$$

$$M_{K \times l} \cdot N_{l \times b} \Rightarrow \# \text{ of scalar multiplications would be } K \times b \times l$$

Note: Matrix multiplication is associative.

Ex: $A(BC) = (AB)C$ for A, B, C matrices with appropriate dimensions

Consider following:

$$A_{n \times n} \cdot B_{n \times n} \cdot X_{n \times 1}$$

$$(A_{n \times n} \cdot B_{n \times n})_{n \times n} \cdot X_{n \times 1} \Rightarrow \begin{matrix} n \times n \times n \\ + \\ n \times 1 \times n \\ = n^3 + n^2 \end{matrix}$$

$$A_{n \times n} \cdot (B_{n \times n} \cdot X_{n \times 1})_{n \times 1} \Rightarrow \begin{matrix} n \times n + n \times n \\ = n^2 + n^2 \\ = 2n^2 \end{matrix}$$

$$n \times 1 \times n$$

$$n \times 1 \times n$$

Rephrase problem:

Parenthesize the product $M_1 M_2 \dots M_n$ in a way to minimize the # of scalar multiplications.

$$M_1 M_2 M_3 \dots M_n$$

Ex: $M_1 = 20 \times 10$

$$M_2 = 10 \times 50$$

$$M_3 = 50 \times 5$$

$$M_4 = 5 \times 30$$

$$M_1 \cdot M_2 \cdot M_3 \cdot M_4$$

$$\left(\underbrace{(M_1 M_2)}_{20 \times 50 \times 10} \underbrace{(M_3 M_4)}_{50 \times 30 \times 5} \right) = 47500$$

$$M_{12} \cdot M_{34}$$

$$\underbrace{\hspace{10em}}$$

$$20 \times 30 \times 50 = 30000$$

$$M_{12} \quad 20 \times 50$$

$$M_{34} \quad 50 \times 30$$

$$M_1 = 20 \times 10$$

$$M_2 = 10 \times 50$$

$$M_3 = 50 \times 5$$

$$M_4 = 5 \times 30$$

$$(M_1 (M_2 M_3) M_4)$$

$$M_2 M_3 = 10 \times 5 \times 30 = 2500$$

$$M_1 \cdot (M_2 M_3)_{10 \times 5} = 20 \times 5 \times 10 = 1000$$

$$(M_1 \cdot (M_2 M_3))_{20 \times 5} \cdot M_4 = 20 \times 30 \times 5 = 3000$$

$$6500$$

$$(M_1 M_2 M_3 \dots M_k) \underset{\uparrow}{(M_{k+1} \dots M_n)}$$

$P(n) = \#$ of alternative parenthesis-
ations of n matrices

$$P(k)$$

$$P(n-k)$$

$$P(n) = \begin{cases} 1 & \text{if } n=1 \\ \sum_{k=1}^{n-1} P(k) P(n-k), & \text{if } n \geq 2 \end{cases}$$

$P(n)$ - n^{th} catalan number

$$P(n) = \frac{1}{n} \binom{2(n-1)}{n-1} \geq \frac{4^{n-1}}{2n^2-n} = \Omega\left(\frac{4^n}{n^2}\right)$$

$$n=10; P(n) \geq \Omega\left(\frac{4^{10}}{10^2}\right) \sim 10^4$$

$$n=20; P(n) \geq \Omega\left(\frac{4^{20}}{20^2}\right) \sim 10^9$$

Dynamic Programming Solution

- Define the table m

- $m[i, j]$ is the number of multiplications performed using optimal parenthesization of

$$M_1 M_2 M_3 \dots \boxed{M_i M_{i+1} \dots M_j} M_{j+1} \dots M_n$$

$$\begin{array}{ccccccc} M_i & M_{i+1} & \dots & M_k & M_{k+1} & \dots & M_j \\ \downarrow & & & \downarrow & \downarrow & & \downarrow \\ d_{i-1} \times d_i & & & d_{k-1} \times d_k & d_k \times d_{k+1} & & d_{j-1} \times d_j \end{array}$$

$$\begin{cases} m[i, j] = \min_k \{ m[i, k] + m[k+1, j] + \\ m[i, i] = 0 \end{cases} \quad d_{i-1} \times d_k \times d_j$$

$m[1, n]$ contains the optimal number of scalar multiplications for the product.

$$M_1 = 20 \times 10$$

$$M_2 = 10 \times 50$$

$$M_3 = 50 \times 5$$

$$M_4 = 5 \times 30$$

$i \backslash j$	1	2	3	4
1	0	10000		
2	-	0	2500	
3	-	-	0	7500
4	-	-	-	0

$$m[1,2] = \min_k \{ m[1,k] + m[k+1,2] + d_0 \times d_k \times d_2 \}$$

$$k=1$$

$$m[1,1] + m[2,2] +$$

$$20 \times 10 \times 50$$

$\approx$

$$m[2,3] = \min_k \{ m[2,k] + m[k+1,3] + \bigcirc \}$$

$$k = 2$$

$$\underbrace{m[2,2]}_0 + \underbrace{m[3,3]}_{d_1 \times d_2 \times d_3} + 10 \times 50 \times 5$$

$$m[3,4] = m[3,3] + m[4,4] + 50 \times 5 \times 30$$

$$= 7500$$

$i \backslash j$	1	2	3	4
1	0	10000	3500	6500
2		0	2500	4000
3			0	7500
4				0

$$M_1 = 20 \times 10$$

$$M_2 = 10 \times 50$$

$$M_3 = 50 \times 5$$

$$M_4 = 5 \times 30$$

$$m[2,4] = \min$$

$$k=2$$

$$m[2,2] + m[3,4] + d_{i-1} \times d_k \times d_j$$

$$0 + 7500$$

$$+ 10 \times 50 \times 30$$

$$7500 + 15000$$

$$k=3$$

$$m[2,3] + m[4,4] + d_{i-1} \times d_k \times d_j$$

$$2500 + 0$$

$$+ 10 \times 5 \times 30$$

$$2500 + 1500$$

$$4000$$

$$m[1,3] = \min$$

$$k=1$$

$$m[1,1] + m[2,3]$$

$$+ d_{i-1} \times d_k \times d_j$$

$$0 + 2500 + 20 \times 5 \times 10$$

$$3500$$

$$k=2$$

$$m[1,2] + m[3,3] +$$

$$d_{i-1} \times d_k \times d_j$$

$$10000 + 0 + 20 \times 5 \times 50$$

$$15000$$

$$m[1,4] = \min$$

$$k=1$$

$$m[1,1] + m[2,4] + d_{i-1} \times d_k \times d_j$$

$$0 + 4000 + 20 \times 10 \times 30$$

$$10000$$

$$k=2$$

$$m[1,2] + m[3,4] + d_{i-1} \times d_k \times d_j$$

$$10000 + 7500 + 20 \times 50 \times 30$$

$$47500$$

$$k=3$$

$$m[1,3] + m[4,4] + d_{i-1} \times d_k \times d_j$$

$$3500 + 0 + 20 \times 5 \times 30$$

$$6500$$